CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc (Biological Science) Examination March, 2018

BS306 Principles of Bioanalysis

Date: 06/04/18 Day: Friday Time: 10.00 a.m. To 12.00 noon Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	**Choose the correct answer for the following questions.**	**10**
1.	ELISA technique is used to detect	
	(i) DNA (ii) Ag-Ab complex (iii) RNA (iv) None of these	
2.	Biomolecules can be separated on the basis of molecular weight using _________ chromatography (i) ion-exchange (ii) affinity (iii) gel filtration (iv) None of these	
3.	DNA absorbs light at _________ nm wavelength.	
	(i) 260 (ii) 270 (iii) 290 (iv) None of these	
4.	rpm stands for ________ per minute.	
	(i) revolutions (ii) rotations (iii) both i & ii (iv) none of these	
5.	Confidentiality of the technician should be maintained during collection and processing of biological samples. (True/False)	

6.	Quantitative analysis provides information about the concentration of test analyte. (True/False)	
7.	Gravimetric analysis measures the amount of an analyte on the basis of ______. (i) mass (ii) charge (iii) volume (iv) none of these	
8.	Two phases are observed during any chromatographic separation. One phase is known as stationary phase and another is known as _____ (i) equivalence (ii) mobile (iii) partition (iv) none of these	
9.	One of the following salts is effectively used for protein precipitation from mixture.	
	(i) ammonium chloride (ii) sodium sulphate (iii) ammonium sulphate (iv) sodium chloride	
10.	Fixed angle rotor of the centrifuge rotor has variable angle during rotation. (True/False)	

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Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

	SECTION -I	
Q – II Answer the following questions as directed		**10**
	Answer **any five** of the following:	
1.	What is the principle of density gradient centrifugation?	
2.	Give definitions of : a) precision and b) accuracy	
3.	Write a brief note on 'salt precipitation of proteins'.	
4	What is RIA? Explain the principle of RIA.	
5.	Draw a well labeled diagram of a pH meter.	
6.	Name different cell and Tissue disruption methods used for protein extraction and briefly explain their principle.	
	SECTION - II	
Q-III Answer the following questions as directed		**15**
	Answer **any three** of the following:	
1.	Explain various types of sampling methods with appropriate examples. What safety measures should be taken during handling of biological samples? Discuss.	
2.	Describe the principle and method of gel filtration chromatography technique.	
3.	How should a standard lab notebook be prepared? Write standard procedure in detail.	
4	Write a brief note on type of audits. Discuss the 'Audit cycle' explaining the importance of steps like audit preparation, audit process and its execution and the follow-up.	
5.	What is electrophoresis? Elaborate the principle and methodology of agarose gel electrophoresis.	